CE 4011 Static + Modal Desktop MVP Installation Manual

Final Documentation Package

Mohammad Umair Naeem
2416055

Spring Semester 2025–2026

1 Purpose

This manual explains how to set up and run the CE 4011 Static + Modal structural analysis desktop MVP on another computer. The final submitted application is a Python Tkinter desktop program. It supports Static Analysis and Modal Analysis; RSA and THA are not part of the submitted desktop scope.

2 Required Software

Table 1: Software requirements

Software	Notes
Python	Assumption: Python 3.10 or newer. Use a standard CPython installation that includes Tkinter.
Tkinter	Bundled with most Windows and macOS Python installers. Some Linux distributions require a separate `python3-tk` package.
matplotlib	Required for model/result visualization and PNG plot export.
pytest	Required for running the automated validation suite.
Git or ZIP extraction tool	Needed to obtain or unpack the repository folder.

The core numerical solver is designed around the Python standard library. Matplotlib is the required visualization dependency.

3 Repository Folder Setup

Place the repository in a normal user-writable folder. Open a terminal in the repository root, which is the folder containing `README.md`, `src/`, `tests/`, and `data/`.

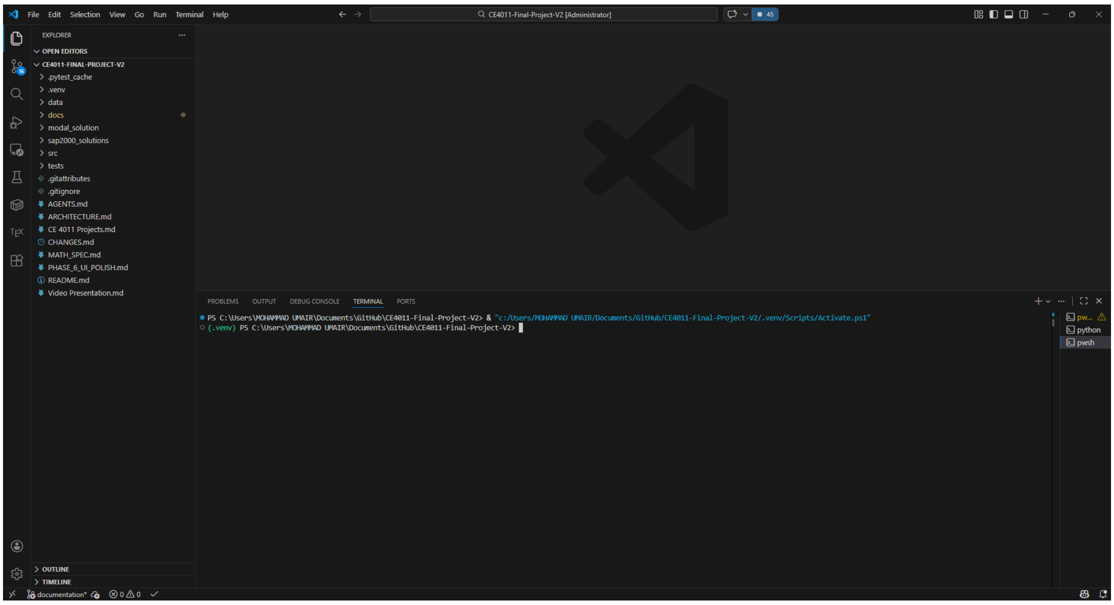

Figure 1: Terminal opened at the repository root after obtaining the project folder.

4 Virtual Environment Setup

On Windows PowerShell:

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install matplotlib pytest
```

On macOS or Linux:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install matplotlib pytest
```

If the command `python` points to the wrong interpreter, use the full path to the intended Python executable.

5 Launch Command

From the repository root, run:

```
python -m src.ui_desktop.app
```

6 Test Command

To run the automated test suite from the repository root:

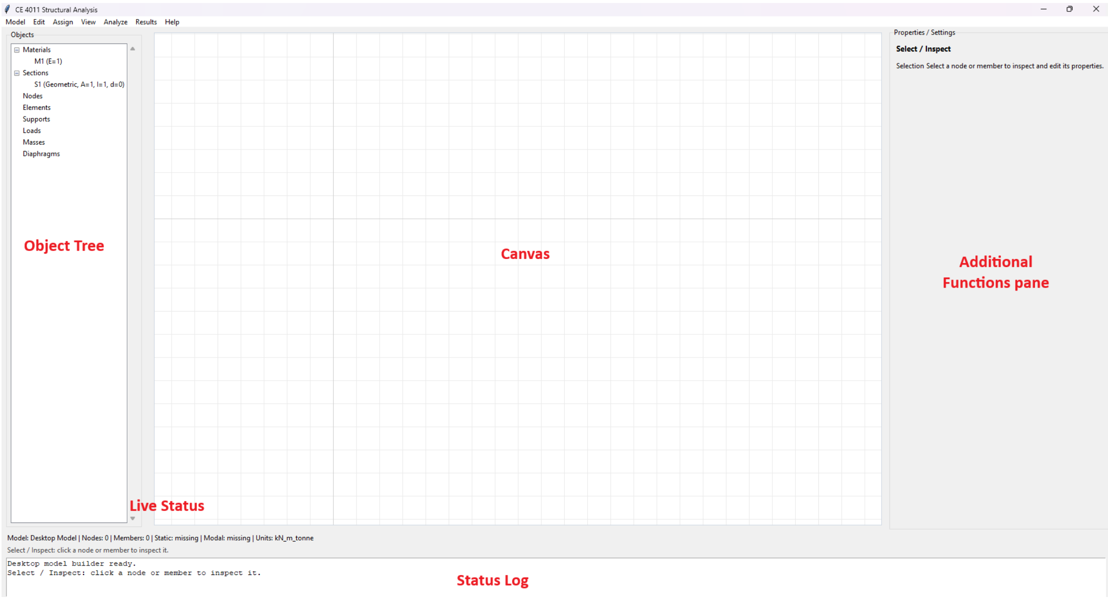

Figure 2: Tkinter desktop application immediately after launch.

```
pytest tests/
```

The final stable baseline documented for submission is `pytest tests/ -q -> 197 passed` in approximately 2 seconds.

7 Troubleshooting

7.1 Wrong Working Directory

Symptom: Python cannot find the `src` module or pytest cannot find tests.

Fix: Change directory to the repository root before running commands. The root contains `src/`, `tests/`, `data/`, and `README.md`.

7.2 Missing Dependencies

Symptom: Import errors for `matplotlib` or `pytest`.

Fix:

```
python -m pip install matplotlib pytest
```

7.3 Tkinter Issues

Symptom: Import errors for `tkinter`, or the desktop window does not open.

Fix: On Windows, reinstall Python using the official installer and keep Tcl/Tk enabled. On Linux, install the distribution Tkinter package, commonly `python3-tk`, then rerun the launch command.

7.4 matplotlib Issues

Symptom: Plot windows or embedded plots fail to render.

Fix: Reinstall matplotlib in the active virtual environment:

```
python -m pip install --upgrade --force-reinstall matplotlib
```

7.5 pytest Not Found

Symptom: The terminal reports that `pytest` is not recognized.

Fix: Activate the virtual environment, or run pytest through Python:

```
python -m pytest tests/
```

8 Final Scope Reminder

Use the desktop application to create or load a model, assign supports/properties/loads/masses, validate the model, run Static or Modal analysis, inspect result tables and plots, and export visible result data where implemented. RSA and THA are future-extension work only.